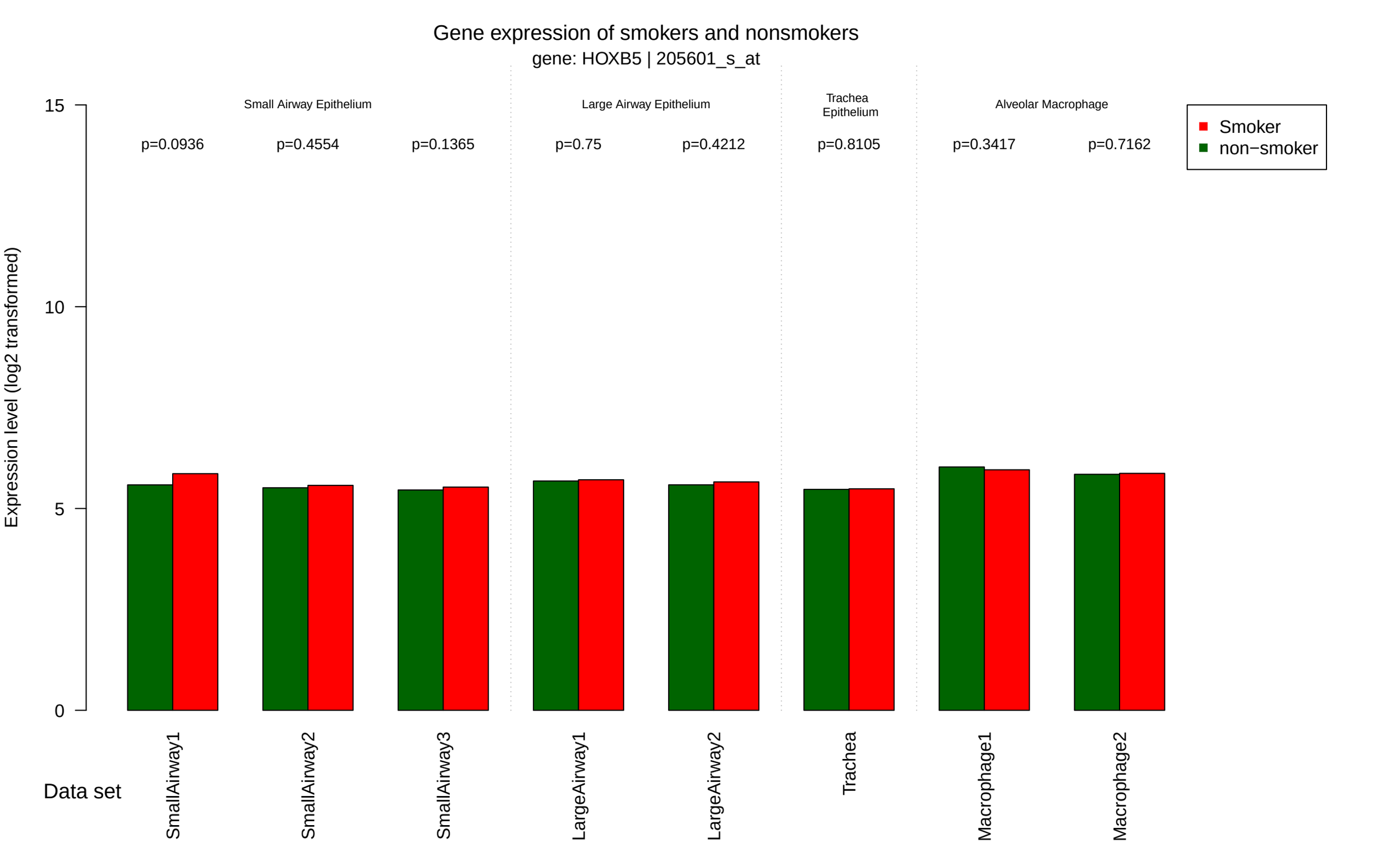
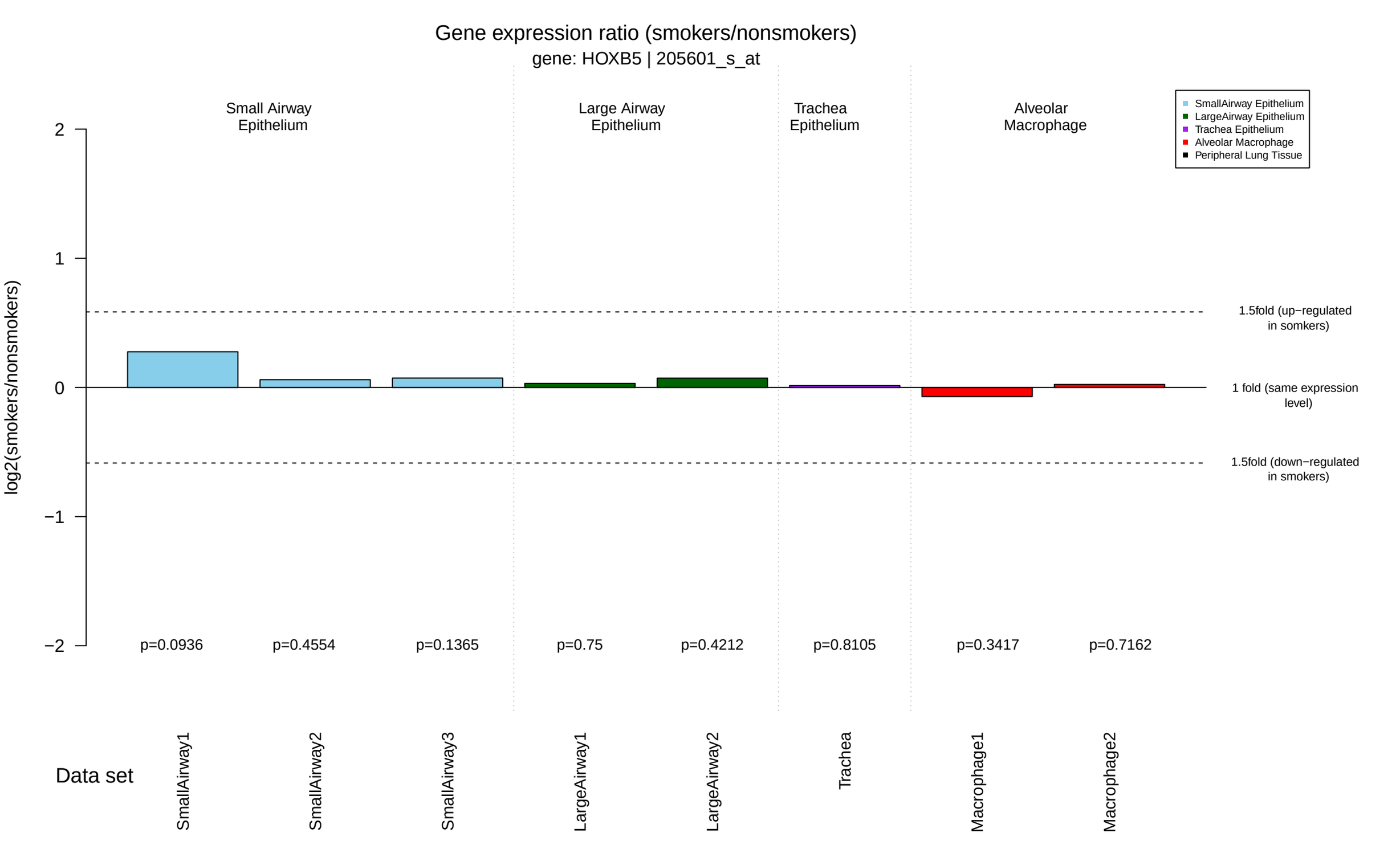
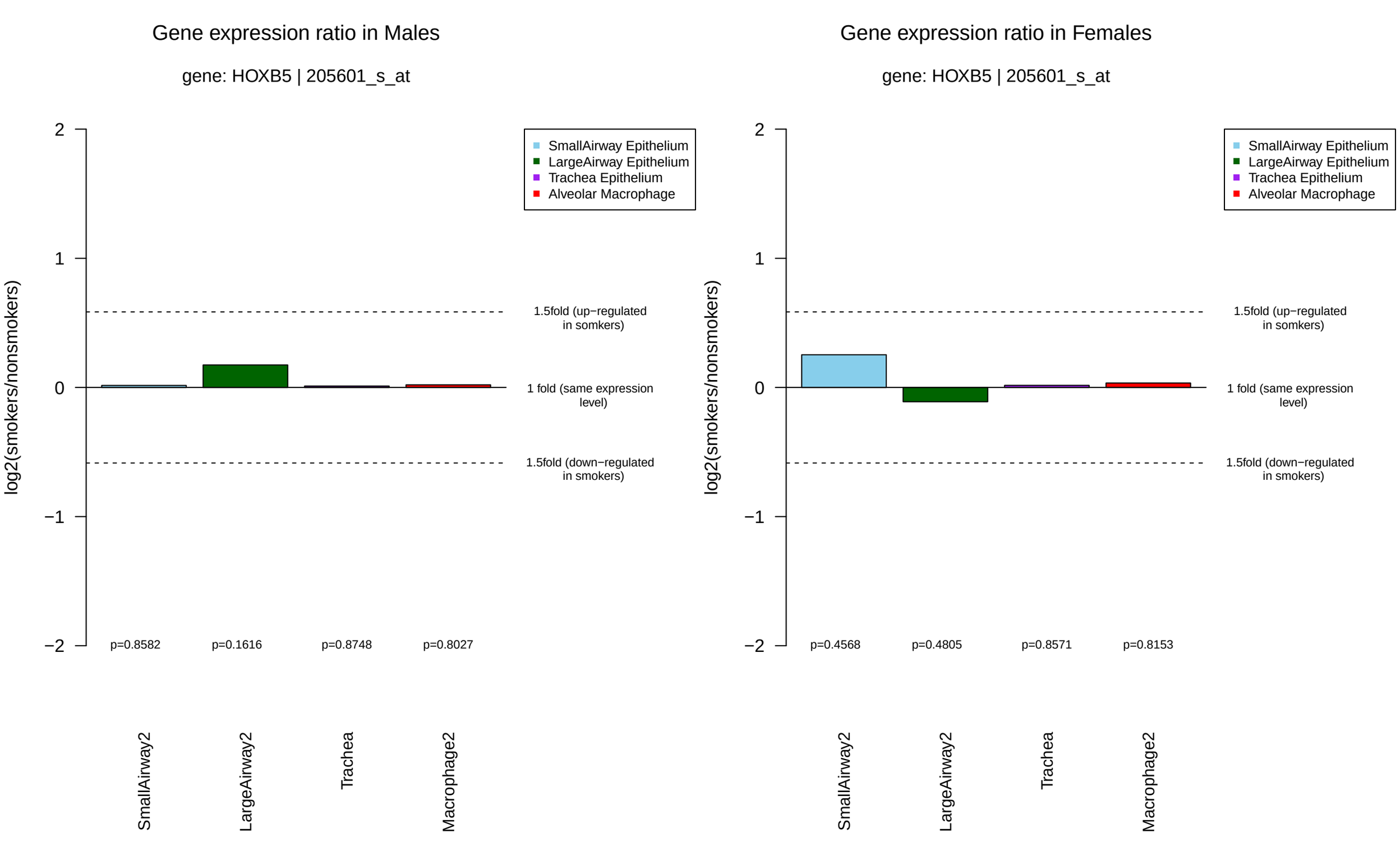
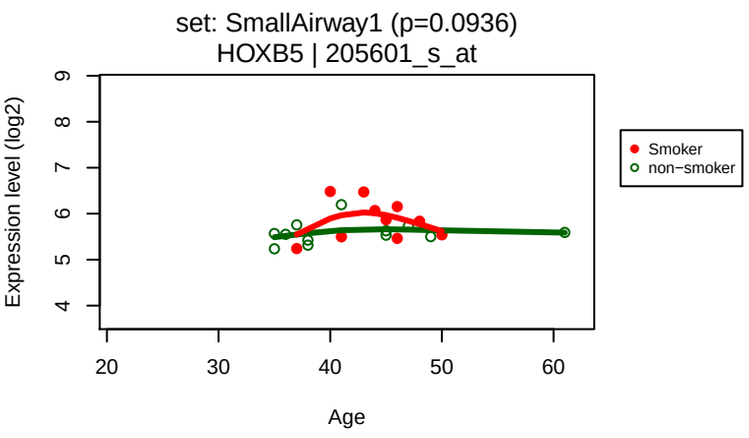
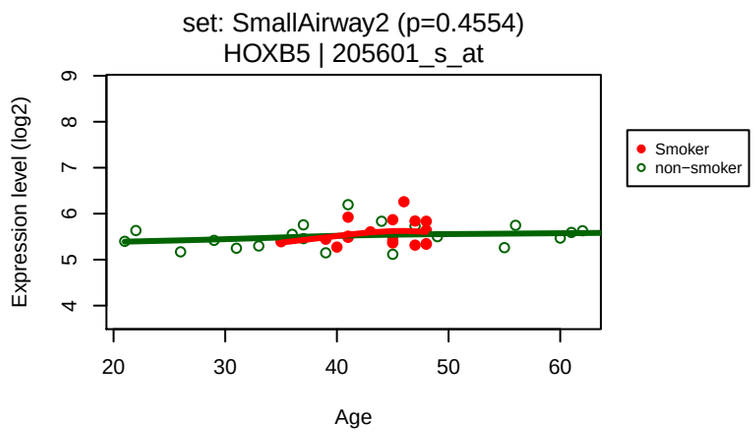
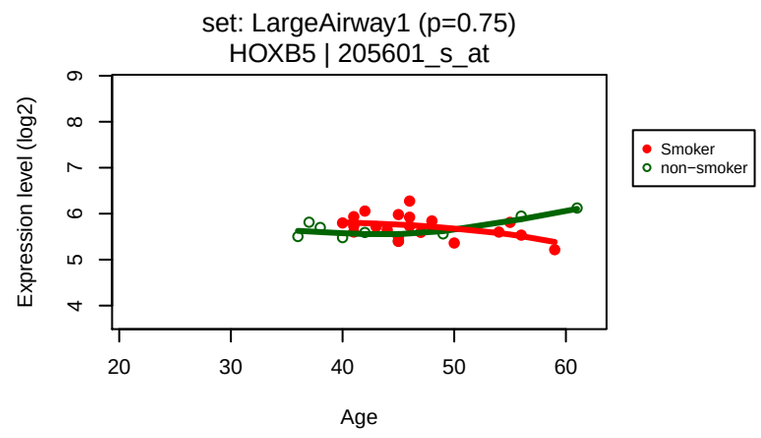

Gene expression of smokers and nonsmokers
gene: HOXB5 | 205601_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0936
p=0.4554
p=0.1365
p=0.75
p=0.4212
p=0.8105
p=0.3417
p=0.7162
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

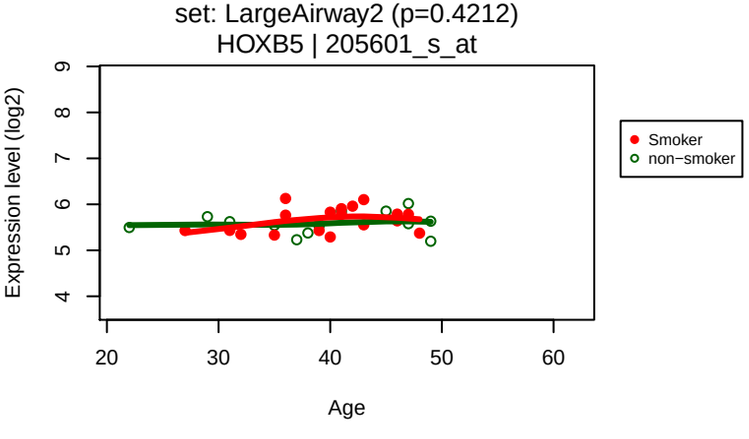

Gene expression ratio (smokers/nonsmokers)
gene: HOXB5 | 205601_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0936
p=0.4554
p=0.1365
p=0.75
p=0.4212
p=0.8105
p=0.3417
p=0.7162
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

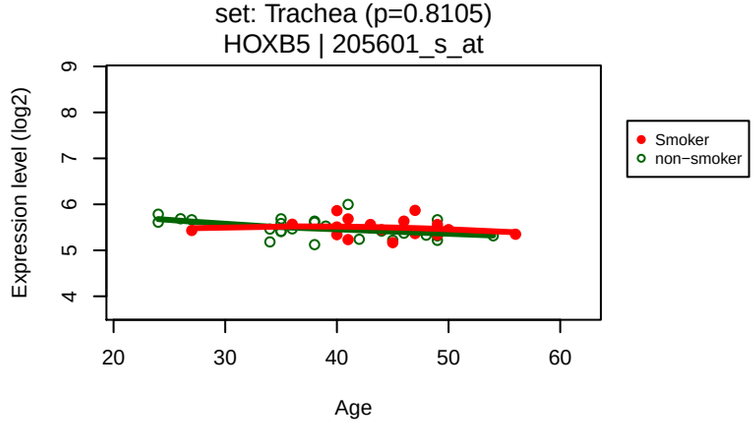

Gene expression ratio in Males
gene: HOXB5 | 205601_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8582
p=0.1616
p=0.8748
p=0.8027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB5 | 205601_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4568
p=0.4805
p=0.8571
p=0.8153
SmallAirway2
LargeAirway2
Trachea
Macrophage2

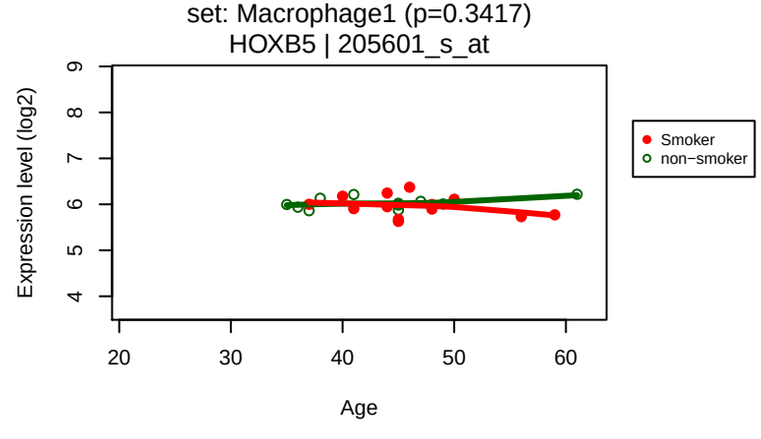

set: SmallAirway1 (p=0.0936)
HOXB5 | 205601_s_at
Expression level (log2)
Age
Smoker
non−smoker

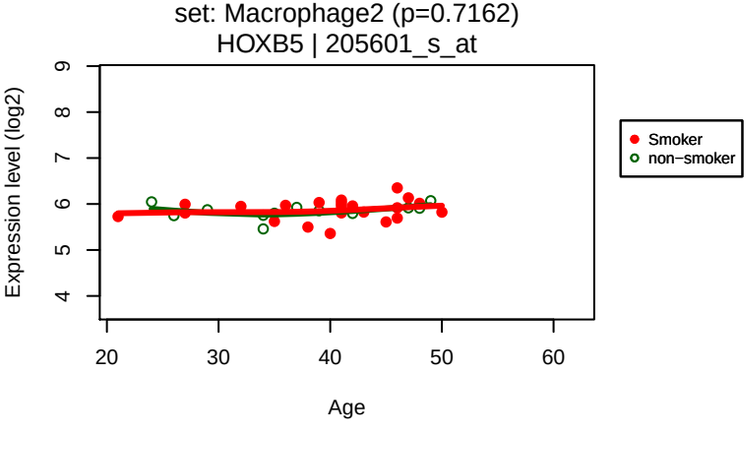

set: SmallAirway2 (p=0.4554)
HOXB5 | 205601_s_at
Expression level (log2)
Age
Smoker
non−smoker

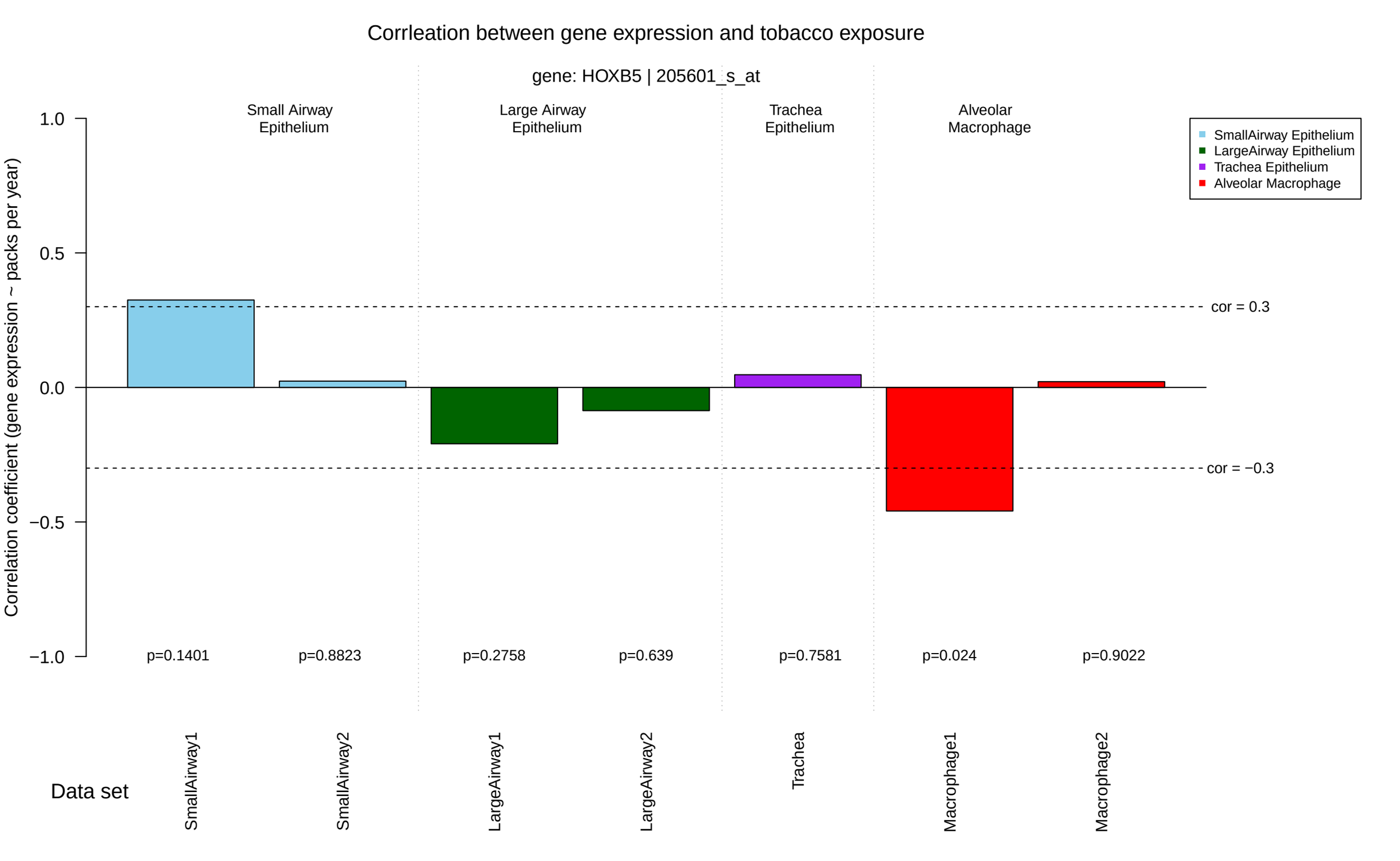

set: LargeAirway1 (p=0.75)
HOXB5 | 205601_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4212)
HOXB5 | 205601_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8105)
HOXB5 | 205601_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3417)
HOXB5 | 205601_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7162)
HOXB5 | 205601_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB5 | 205601_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1401
p=0.8823
p=0.2758
p=0.639
p=0.7581
p=0.024
p=0.9022
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2